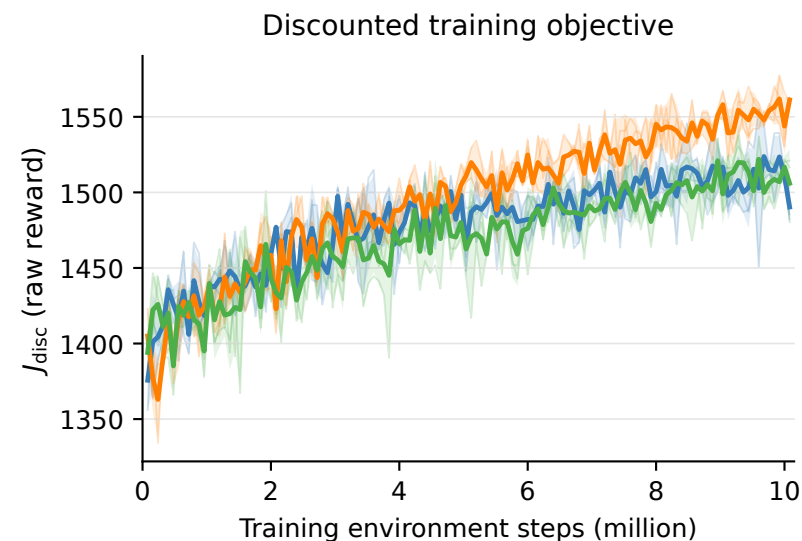
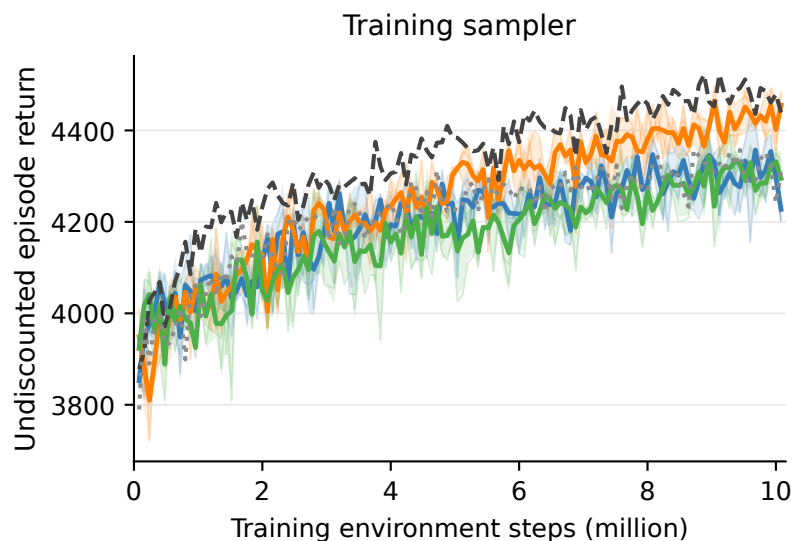
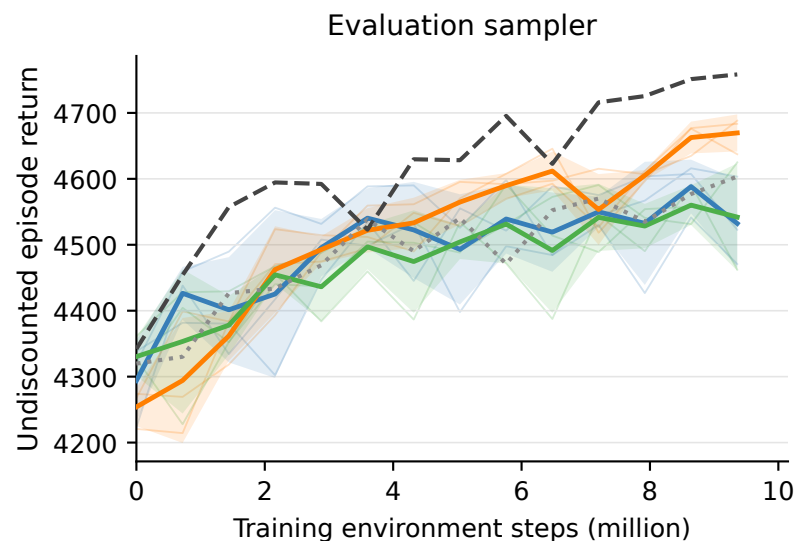


Stage 3: performance across completed wave 1

Nine conditions · seeds 0-2 · mean $\pm$ sample SD · faint individual seeds · no smoothing

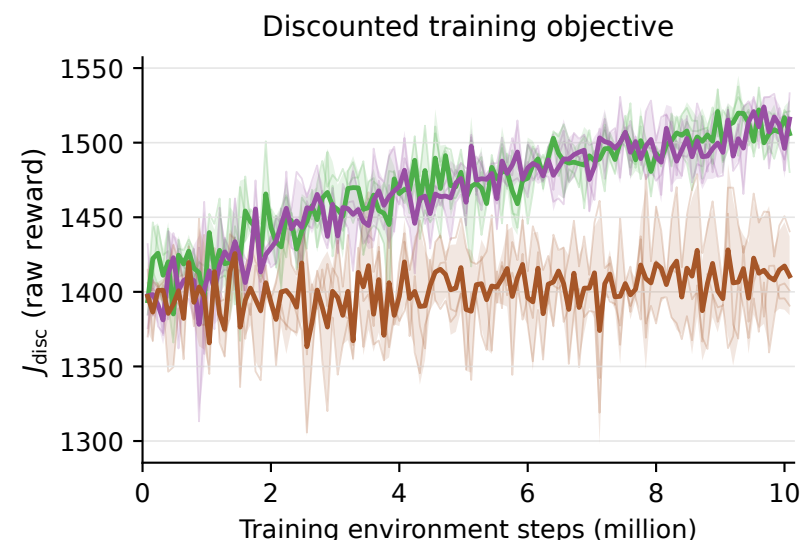
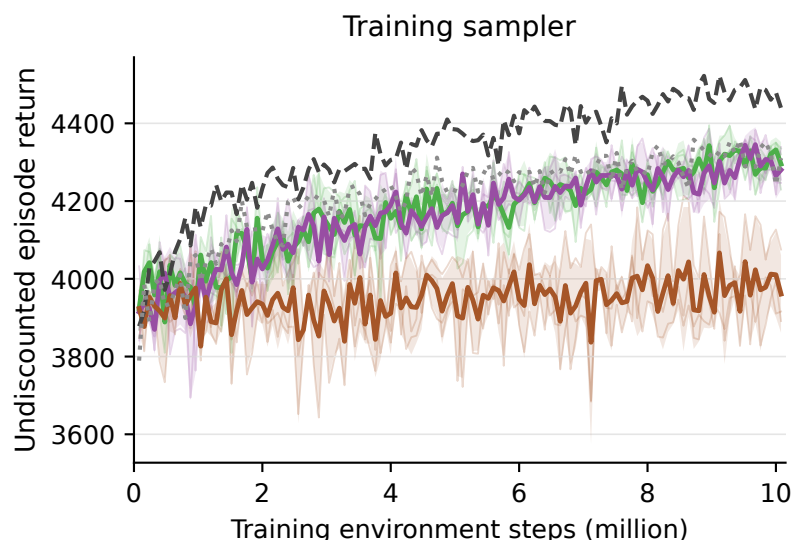
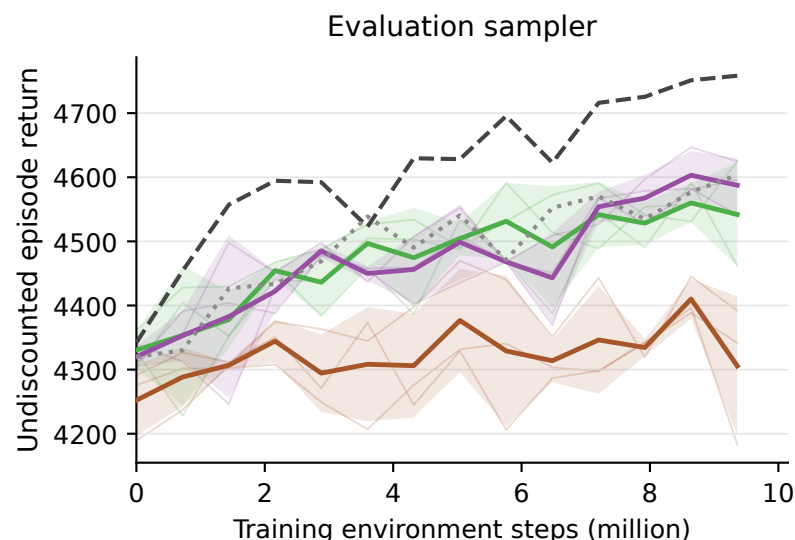
Estimator ladder under AdamW

E0 E1 E2 --- Stage 1 DPPO Stage 2 NC4 1e-4



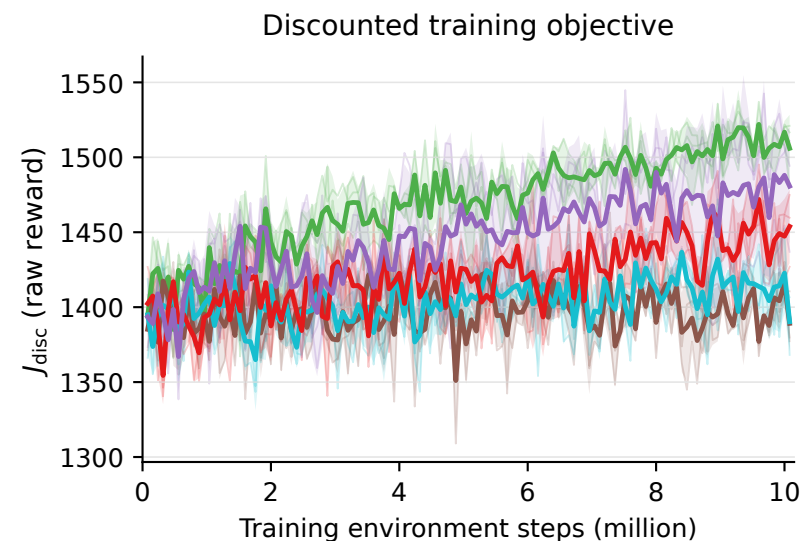
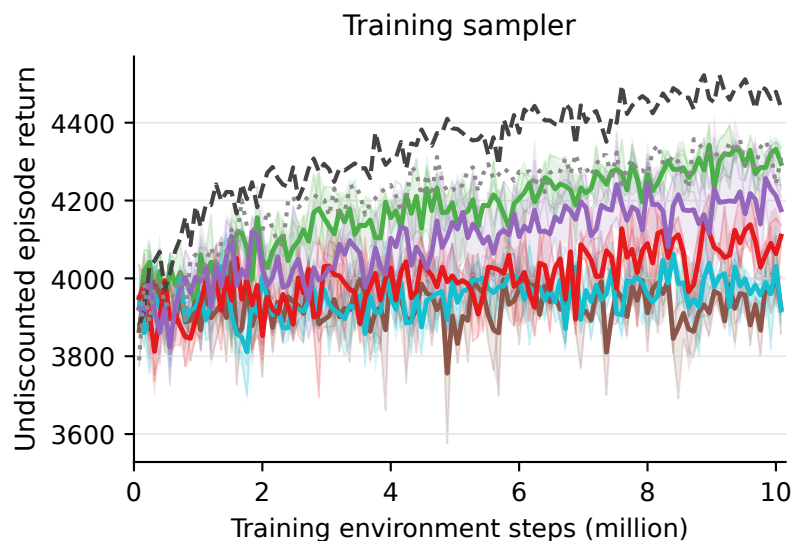
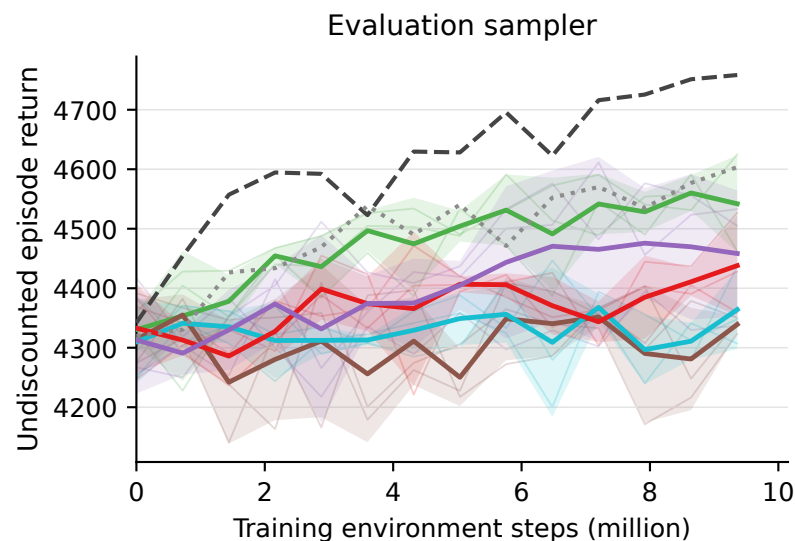
E2 under AdamW: momentum and baseline

E2 E2_beta0 E2_nobase --- Stage 1 DPPO Stage 2 NC4 1e-4



E2 estimator: AdamW and the SGD step-size sweep

E2 SGD03 SGD1 SGD3 SGD10 --- Stage 1 DPPO Stage 2 NC4 1e-4



Evaluation ends at 9.36M steps; training at 10.08M. Gray curves: earlier-stage means where recorded.